SUMLINK

Sumlink Analytics Dashboard

Production Implementation and Validation Report

Report item	Details
Prepared for	Management Review
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Report date	03 August 2026
GA4 property	516899630
Dashboard	Appsmith Cloud
Backend	FastAPI on Render

Confidentiality: Internal project report. Credentials, API keys, and service-account secrets are intentionally excluded.

1. Executive Summary

The Sumlink Analytics Dashboard is a production-ready product analytics solution for management and game-performance review. It combines an Appsmith Cloud interface with a FastAPI backend hosted on Render. The backend retrieves aggregated reporting data from the Google Analytics 4 Data API and exact user-level rolling-retention data from the GA4 BigQuery export.

The delivered dashboard contains five management sections: Executive Health, Acquisition, Onboarding, Gameplay, and Retention. The production backend is live, GA4 connectivity is confirmed, exact rolling retention is enabled through BigQuery, and the focused backend validation suite passed all tests on 03 August 2026.

Area	Status	Evidence
Render backend	Operational	/health returned ok; version 1.0.0
GA4 connection	Connected	/ready returned ready and GA4 connected
Appsmith dashboard	Deployed	Five sections imported into Appsmith Cloud
BigQuery retention	Enabled	Exact user-level rolling retention returned 11 cohort rows
Automated tests	Passed	5 focused API and BigQuery tests passed

2. Project Objective

The objective is to give management one consistent view of product health, user acquisition, onboarding quality, gameplay balance, and retention. The dashboard replaces fragmented manual checks with reusable filters, documented KPI definitions, and API-backed reporting.

- Provide a concise management view of user growth, engagement, conversion, gameplay, and retention.
- Use live GA4 reporting data while keeping credentials outside the Appsmith client.
- Support exact rolling-retention analysis using BigQuery user-level events.
- Allow managers to review the deployed dashboard without running a developer laptop.

3. Production Architecture

Layer	Technology	Responsibility
Dashboard	Appsmith Cloud	Filters, KPI cards, charts, tables, and manager-facing navigation
API	Python FastAPI	Authentication, validation, aggregation, caching, and response shaping
Hosting	Render	Public HTTPS backend deployment and health monitoring
Reporting data	GA4 Data API	Aggregated users, sessions, engagement, funnels, events, and cohort metrics
Raw analytics	GA4 BigQuery export	Exact distinct-user rolling retention and user-level cohort calculation

Data flow: Manager -> Appsmith Cloud -> HTTPS FastAPI endpoint -> GA4 Data API and/or BigQuery -> normalized JSON response -> dashboard cards, charts, and tables.

Appsmith never receives the Google service-account credential. It sends only dashboard filters and the protected API header to the Render backend.

4. Hosting and Deployment

Component	Production location	Current state
Backend API	https://sumlink-analytics-api.onrender.com	Live and externally reachable
Health check	/health	Status ok; service version 1.0.0
Readiness check	/ready	Status ready; GA4 connected
Dashboard	xorstack.appsmith.com workspace	Imported and deployed in Appsmith Cloud
Local environment	localhost	Development copy only; not required for production access

Manager access must use the deployed Appsmith Cloud application URL. If login is required, invite the manager through Appsmith Share or make the application public according to company policy.

5. Dashboard Modules

5.1 Executive Health

- Active users and new users for the selected period.
- DAU, MAU, and DAU/MAU stickiness.
- Engagement rate, sessions, screen views, and activity trends.
- A compact management view of product adoption and engagement.

5.2 Acquisition

- User acquisition and churn-oriented summary metrics.
- Channel, source, and campaign quality comparisons where GA4 dimensions are available.
- New-user growth and returning-user behavior for the selected filters.

5.3 Onboarding

- Onboarding and tutorial funnel progression.
- Tutorial started, completed, failed, skipped, and frustration indicators.
- Conversion and drop-off rates across onboarding steps.
- Average tutorial-step timing where registered GA4 parameters are available.

5.4 Gameplay

- Level starts, completions, failures, and drop-off analysis.

- New versus returning player activity.
- Hint and add-row usage, level difficulty, and completion trends.
- Level-level tables for identifying balancing and progression issues.

5.5 Retention

- DAU, WAU, MAU, stickiness, sessions, and returning users.
- Exact Day 1, Day 3, Day 7, and Day 15 cohort retention from GA4 reporting.
- Exact rolling Day 1+, Day 3+, Day 7+, and Day 15+ retention from BigQuery.
- Cohort tables and trends with mature-day handling for incomplete cohorts.

6. Global Filters

Filter	Purpose
Date range	Last 7, 14, or 30 days and supported custom date ranges
App version	Compare releases and identify version-specific changes
OS version	Segment platform behavior
Device model	Identify device-specific differences
Country	Review geographic performance
Level	Focus gameplay metrics on a selected level

7. KPI Definitions and Formulas

KPI	Definition or formula
Engagement rate	Engaged sessions / Sessions x 100
Stickiness	DAU / MAU x 100
Onboarding completion	Users completing onboarding / Users starting onboarding x 100
Level completion	Completed level events / Started level events x 100
Level drop-off	(Level starts - Level completions) / Level starts x 100
Failure rate	Failed events / Started events x 100
Skip rate	Skipped events / Started events x 100
Exact Day N retention	Day 0 cohort users active exactly on Day N / Day 0 cohort users x 100
Rolling Day N+ retention	Day 0 cohort users active on Day N or any later day / Day 0 cohort users x 100

8. GA4 and BigQuery Data Methodology

8.1 GA4 Data API

The GA4 Data API supplies aggregated reporting including active users, new users, sessions, engaged sessions, event counts, registered custom dimensions, registered custom metrics, funnels, filters, and standard cohort-style metrics. It follows GA4 reporting definitions and privacy controls.

8.2 BigQuery GA4 Export

BigQuery is used where exact user-level event history is required. The retention service scans the GA4 events_* export, derives each user's Day 0 cohort from first activity, calculates later activity offsets, and counts distinct user_pseudo_id values. This produces exact rolling retention rather than an aggregate estimate.

8.3 Rolling Retention

Rolling Day N+ retention measures whether a user from a Day 0 cohort returned on Day N or any later day. A cohort day is displayed only when enough time has elapsed to mature. The All Users result is a weighted aggregate of eligible cohorts, preventing small cohorts from distorting the overall percentage.

9. Validation Results

Validation timestamp: 03 August 2026, 23:58 IST. These values were refreshed from the live Render service immediately before this report was generated.

Live measure	Verified value
GA4 connection	Connected
Active users yesterday	34
Active users - selected range	602
New / Day 0 cohort users	141
DAU / WAU / MAU	34 / 160 / 549
Stickiness	6.19%
Sessions / Engaged sessions	2,861 / 2,169
Day 1 retention	18.44% (26 users)
Day 3 retention	4.96% (7 users)
Day 7 retention	0.71% (1 user)
Day 15 retention	0.71% (1 user)
Rolling cohort rows	11
Rolling data source	BigQuery exact user-level

Automated verification	Result
BigQuery retention service tests	Passed
API response tests	Passed
Focused test total	5 passed in 2.94 seconds
Render health check	Passed
Render readiness and GA4 connection	Passed

Validation conclusion: Standard Day N retention uses GA4 cohort reporting, while rolling Day N+ retention uses exact distinct-user calculations from the BigQuery export.

10. API Interface

Dashboard section	Production API path
Executive Health	/api/v1/executive/summary
Acquisition	/api/v1/acquisition/summary
Onboarding	/api/v1/onboarding/summary
Gameplay	/api/v1/gameplay/summary
Retention	/api/v1/retention/summary

Protected dashboard requests use the x-api-key header. The key value is intentionally not included in this report.

11. Security and Operational Controls

- API authentication is enabled for protected production routes.

- Google service-account credentials are stored as backend environment secrets.
- The Appsmith interface does not contain or expose Google credentials.
- CORS is configured for the Appsmith Cloud origin.
- Backend responses are cached for 300 seconds to reduce repeated GA4 queries.
- Production readiness is monitored through dedicated health and readiness routes.

12. Limitations and Operational Notes

- The Render free service can sleep after inactivity; the first request may require a short wake-up period.
- GA4 Data API results are aggregated and can differ from raw-event queries when definitions, identity, filters, time zones, or processing windows differ.
- BigQuery rolling retention depends on the availability and completeness of the GA4 events_* export.
- Recent cohorts correctly show unavailable values for retention days that have not yet matured.
- Manager access depends on Appsmith sharing configuration: public application or invited workspace user.
- Some historical dashboard comparison labels showed character-encoding artifacts; these do not change the validated KPI calculations.

13. Final Project Status

Deliverable	Final status
Five-section analytics dashboard	Complete
FastAPI GA4 backend	Deployed on Render
Appsmith Cloud application	Imported and deployed
GA4 Data API integration	Validated
BigQuery exact rolling retention	Validated
Authentication and production configuration	Enabled
Manager documentation	Complete

14. Manager-Ready Conclusion

The Sumlink Analytics Dashboard is ready for management review. It provides one consolidated view of executive health, acquisition, onboarding, gameplay, and retention. The production backend is live on Render, the dashboard is deployed in Appsmith Cloud, GA4 connectivity is confirmed, and exact rolling-retention calculations are supported by BigQuery user-level data.

The manager should receive the deployed Appsmith Cloud application link rather than a localhost URL. Before sharing, confirm access in a private browser window and either enable approved public access or invite the manager's email address.